# Thinesh Thiyakesan Ponbagavathi

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## Education

**University of Stuttgart, Germany** Jan 2024 – Oct 2027 (Expected)

PhD in Computer Science advised by Prof. Dr. Alina Roitberg
Scholar at the International Max Planck Research School for Intelligent Systems (IMPRS-IS)

**University of Stuttgart, Germany** 2021 – 2023

M.Sc. (1.6/1.0) in Electrical Engineering (Smart Systems)
*Thesis:* 3D-Aware Urban Scene Generation advised by Prof. Dr. Bin Yang

**Anna University, India** 2014 – 2018

B.E. (8.63/10) in Electronics and Instrumentation Engineering
*Thesis:* Cognitive Smart Cities using LoRa advised by Prof. Dr. C. Esakkiappan

## Research Interests

Computer Vision · Vision Foundation Models · Video Understanding · Parameter-Efficient Fine-Tuning (PEFT) · View-Invariant Representation Learning · Human-Robot Interaction

## Experience

**Applied Scientist Intern**, **Amazon, USA** Aug 2026 – Present

Building a framework to measure memorization vs. generalization in Amazon's Delivery Foundation Model.
Pretraining models up to 0.5B parameters across regions and data scales to study geographic scaling laws.

**PhD Researcher**, **University of Stuttgart & University of Hildesheim, Germany** 2024 – Present

Develop parameter-efficient adaptations of vision foundation models for fine-grained video understanding.
Design cross-view and temporal modeling strategies for robust recognition of nearly symmetric actions.

**Lead Controls Engineer**, **Johnson Controls, India** 2018 – 2021

Designed control systems for HVAC and building automation across large-scale projects.

## Selected Publications

- *Not All Layers are Equal for Image-to-Video Transfer.*
  **Thinesh Thiyakesan Ponbagavathi**, Constantin Seibold and Alina Roitberg
  *Advances in Neural Information Processing Systems* (***NeurIPS***), 2026.

- *Frame2Freq: Spectral Adapters for Fine-Grained Video Understanding.*
  **Thinesh Thiyakesan Ponbagavathi**, Constantin Seibold and Alina Roitberg
  *IEEE/CVF Conf. on Computer Vision and Pattern Recognition* (***CVPR***), 2026.

- *Order Matters: On Parameter-Efficient Image-to-Video Probing for Recognizing Nearly Symmetric Actions.*
  **Thinesh Thiyakesan Ponbagavathi** and Alina Roitberg
  *IEEE Int. Conf. on Robotics and Automation* (***ICRA***), 2026.

- *Structured Relational Reasoning for Group Activity Assessment.*
  **Thinesh Thiyakesan Ponbagavathi***, Chengzheng Yang* and Alina Roitberg (*Equal contribution)
  *IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops* (***CVPRW***), 2026.

- *T-MASK: Temporal Masking for Probing Foundation Models across Camera Views in Driver Monitoring.*
  **Thinesh Thiyakesan Ponbagavathi**, Kunyu Peng and Alina Roitberg
  *IEEE Int. Conf. on Intelligent Transportation Systems* (***ITSC***), 2025. ***Best Student Paper Finalist.***

## Academic Activities

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| Selected Participant, International Computer Vision Summer School (ICVSS), Sicily | 2026 |
| Supervisor for (5×) M.Sc. Theses | 2024 – Present |
| Teaching Assistant, Automotive and Assistive Computer Vision | 2025 |
| Oral Presentation, High Performance Computing in Science & Engineering, HLRS Workshop | 2025 |
| Lead Organizer, Seminar on Limitations of Foundation Models, Hildesheim | 2026 – Present |
| Lead Organizer, Lab Course on Multimodal Recognition & Deep Learning, Hildesheim | 2026 – Present |
| Reviewer: IEEE IV (2025), ITSC (2025), ICRA (2026), ECCVW(2026), GCPR (2026) | 2025 – 2026 |

## Skills

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| **Programming Languages:** | Python, C++ |
| **Frameworks & Libraries:** | PyTorch, TensorFlow, HuggingFace, OpenCV |
| **Tools:** | Git, SLURM, LaTeX, Distributed Training (DDP) |
| **Languages:** | Tamil (Native), English (Full Professional Proficiency), German (Elementary) |

## Awards and Honors

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|---|---|
| Best Student Paper Finalist, IEEE Int. Conf. on Intelligent Transportation Systems (ITSC) | 2025 |
| PhD Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS) | 2024 |
| Rank 2/22, M.Sc. Electrical Engineering, University of Stuttgart | 2023 |
| University Rank Medal, Anna University (Rank 18/1079) | 2018 |
| Department Topper Cash Prize (3×) for topping semester examinations, Anna University | 2014–2018 |

## Additional Publications

- *PDNet: Exploring Pruning and Distillation for Real-Time Worker Activity Recognition in Industrial Assembly.*
  Malek Baba, **Thinesh Thiyakesan Ponbagavathi** and Alina Roitberg
  *Int. Conf. on Computer Vision Theory and Applications* (***VISAPP***), 2026. ***Oral Presentation.***

- *SceneGraphRAG: Efficient and Scalable Object Grounding Using Knowledge-Enhanced 3D Scene Graphs.*
  Michael Pabst, **Thinesh Thiyakesan Ponbagavathi**, Deniz Bickici, Dennis Rotondi, Shohei Mori, Kai O. Arras and Dieter Schmalstieg
  Paper Under Review, 2026.

- *RealityGraph: Generating, Editing and Applying Semantic Scene Graphs in Augmented Reality.*
  Michael Pabst, **Thinesh Thiyakesan Ponbagavathi**, Jan Kolberg, Verena Biener, Shohei Mori, Alina Roitberg and Dieter Schmalstieg
  Paper Under Review, 2026.